Cloud GPU Training – Manager Discussion Sheet

Satellite imagery: object detection, classification and temporal change detection | Planning case: 1M+ images @ 1024×1024

1. CURRENT BASELINE & SCALE-UP

Parameter	Previous Run	Target / Planning Case	Discussion Point
Training images	150880	1000000	Dataset is now 6.6×+ larger; confirm final usable/deduplicated count.
Training resolution	640×640	1024×1024	2.56× pixels per image; preserve small-object detail via tiling rather than global resize.
Batch size	~4	Fit to GPU; target ~16–32 on L4/L40s	Benchmark maximum stable batch with AMP; do not fix batch arbitrarily.
Epochs	85	50	50 is a planning case. Use early stopping and validation curves; do not assume more epochs = more accuracy.
Observed training time	~90–100 min/epoch	To be benchmarked	Old run also showed ~11 min validation. New runtime should be based on measured images/sec.

2. PROPOSED PILOT INSTANCE

Component	Preferred Request	Minimum / Alternative	Reason
GPU	1× NVIDIA L40S 48 GB	L4 24 GB only for comparison	L40S is the primary training candidate; larger VRAM enables higher batch at 1024.
CPU	32 vCPU requested for benchmark	16 vCPU starting config	Image decoding, augmentation and DataLoader workers must not starve the GPU.
RAM	128 GB preferred	64 GB starting config	Allows caching/prefetching and large worker queues.
Storage	1–2 TB NVMe / high-IOPS block storage	Size after dataset audit	Keep training dataset close to compute; avoid slow network-storage bottlenecks.
OS / stack	Ubuntu + CUDA + PyTorch + Ultralytics/required libs	Root/self-managed access	Environment should be reproducible and controlled by project team.
Access	SSH key + MFA + restrictive NSG/VPN	No password-based public SSH	Required for restricted/private dataset handling.

3. PUBLIC PRICING SNAPSHOT – ACECLOUD, NOIDA (verified 27 Oct 2026)

Item	Public figure	Planning interpretation	Source / verification	Hours	Compute ex-GST	Compute incl. 18% GST
L40S hourly	₹142.12	Primary PAYG benchmark rate; verify quote and exact included resources.	https://acecloud.ai/cloud/gpu/	250	₹35,530	₹41,925
L40S monthly	₹83,000.00	Starting configuration: 1× L40S, 16 vCPU, 64 GB RAM.	https://acecloud.ai/cloud/gpu/	300	₹42,636	₹50,310
GST	18%	Public GPU page states prices exclude 18% GST.	https://acecloud.ai/cloud/gpu/	350	₹49,742	₹58,696
High-IOPS block storage	₹6.00	₹/GB/month in Noida; 1 TiB ≈ ₹6,144/month before GST.	https://acecloud.ai/blog/block-storage-for-databases/	400	₹56,848	₹67,081
Advertised new-customer credit	₹20,000.00	Ask whether credit applies to the benchmark and confirm validity/conditions in writing.	https://acecloud.ai/cloud/gpu/			

4. MANDATORY SECURITY / COMPLIANCE REQUIREMENTS – MUST BE CONFIRMED IN WRITING

P0	Institutional authorization	Written approval that restricted/private imagery may be uploaded to a third-party cloud.	NO DATA UPLOAD BEFORE APPROVAL
P0	Data residency	Exact region/data-centre location; require India-only residency if mandated by owner/institution.	WRITTEN CONFIRMATION
P0	Provider staff access	Document whether provider administrators can access VM/storage; support access must be explicit, time-bound and logged.	WRITTEN ACCESS MODEL
P0	Encryption	Encryption at rest + in transit; clarify who owns/controls encryption keys.	KEY OWNERSHIP + METHOD
P0	Network access	SSH keys, MFA on control plane, NSG/firewall allowlist and preferably VPN/private networking.	NO OPEN PASSWORD SSH
P0	Backups & snapshots	Default behaviour, locations, retention and whether snapshots/backups can be disabled.	NO UNAPPROVED COPIES
P0	Auditability	Admin/support access logs, control-plane logs and retention period.	AUDIT LOG EVIDENCE
P0	Secure deletion	Deletion of VM disks, detached volumes, snapshots and backups at project termination.	DELETION PROCEDURE
P0	Legal / contractual	NDA/DPA, confidentiality, breach notification, subprocessor list and data-handling terms.	DOCUMENTS BEFORE PRODUCTION
P1	Certifications	Request current ISO/IEC 27001, 27017 and 27018 evidence / scope statement.	VERIFY SCOPE, NOT JUST LOGO

5. TARGET OUTCOME FROM MANAGER MEETING

#	Required outcome
1	Benchmark access to 1× L40S with representative 50k–100k image subset before production commitment.
2	Measure images/sec, epoch time, GPU utilization, VRAM usage, CPU usage, storage throughput and cost/hour.
3	Written quote for hourly L40S configuration with 16 vs 32 vCPU, 64 vs 128 GB RAM, and 1–2 TB NVMe/high-IOPS storage.
4	Written security architecture / responsibilities: data residency, provider access, encryption, backups, logging and deletion.
5	Confirm ₹20,000 credit eligibility, validity, billing start/stop semantics, monthly-vs-hourly break-even, and all add-on charges.
6	**Do not upload the real restricted dataset until institutional authorization and required contractual controls are in place.**